

Question				Marking details	Marks available																	
					AO1	AO2	AO3	Total	Maths	Prac												
9.	(a)			Alpha – <u>helium nucleus</u> (1) Treat 2 protons and 2 neutrons as neutral Beta – [fast moving] electron (1) Gamma – em wave (1)	3			3														
	(b)	(i)		$\frac{18}{30} = 0.6$ [counts per second]		1		1	1	1												
		(ii)		Measure for longer (1) Repeat (1)			2	2		2												
		(iii)		[More] radon (1) Due to granite or different types of rock (1) Alternative: [More] cosmic rays (1) Due to high altitude (1)	2			2														
	(c)	(i)		<table border="1"><tr><td></td><td>Alpha</td><td>Beta</td><td>Gamma</td></tr><tr><td>Source 1</td><td>N</td><td>N</td><td>Y</td></tr><tr><td>Source 2</td><td>Y (1)</td><td>Y (1)</td><td>N (1)</td></tr></table> Award 1 mark for source 1 row being correct Accept ticks and crosses for Y and N		Alpha	Beta	Gamma	Source 1	N	N	Y	Source 2	Y (1)	Y (1)	N (1)			4	4		4
	Alpha	Beta	Gamma																			
Source 1	N	N	Y																			
Source 2	Y (1)	Y (1)	N (1)																			

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		(ii)		Lead or concrete [absorber] should be included (1) To confirm that gamma [is emitted] (1) OR Reduce distance between source and detector (1) To detect alpha (1) OR Use thick aluminium (1) To confirm no beta is emitted by source 1 (1)			2	2		2
				Question 9 total	5	1	8	14	1	9